UNIT 3 · TOPIC 1(A)

Calibration of Seed Drills

Syllabus topic: Calibration of seed drills, planters and plant protection equipment

What calibration is

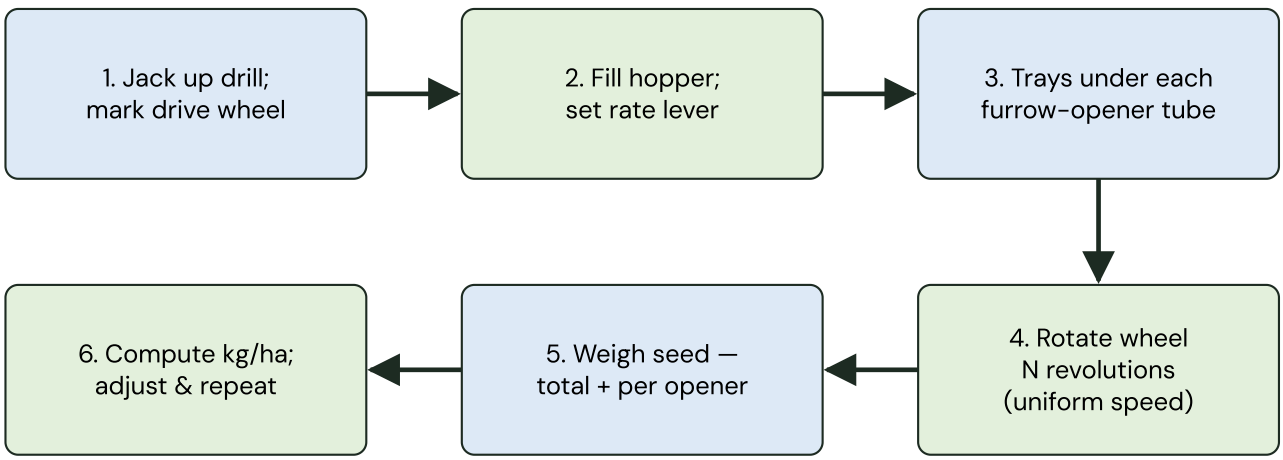
Calibration determines and sets the actual application rate (seed, fertilizer or chemical) of a machine by **simulating field operation** — stationary, or over a measured strip — **before** field work.

- Hit the recommended rate.
- Avoid costly over- or under-application.
- Check uniformity between metering units.
- Verify the machine after a repair or a change of seed lot.

> **Golden rule:** always calibrate with the ACTUAL lot you will sow — seed size, shape, density, treatment coating and moisture change the metered rate even at the same lever setting.

Procedure (ground-wheel-driven drill)

1. Jack up the drill and mark the drive wheel.
2. Fill the hopper and set the rate lever.
3. Place a tray under each furrow-opener tube.
4. Run a few turns first to fill the metering mechanism (do not count these).
5. Rotate the wheel N revolutions at a speed corresponding to field speed.
6. Weigh the seed — total and per opener.
7. Compute kg/ha; adjust and repeat until the rate is within tolerance.



Seed-drill calibration procedure — stationary simulation of field work (redrawn from the source figure)

Key relations

$$\text{Distance per wheel revolution} = \pi D \ (D = \text{drive/ground wheel diameter})$$

$$\text{Working width } W = \text{number of rows} \times \text{row spacing}$$

$$A\ (m^2) = \pi D \times N \times W$$

$$\text{Seed rate (kg/ha)} = 10{,}000 \times \text{weight collected (kg)} \div A\ (m^2)$$

Choose N to represent a convenient area — e.g., for 1/25 ha (400 m²), N = 400/(πDW). Worked numerical examples are in Topic 6b.

Metering mechanisms and rate adjustment

Mechanism	How the rate is changed
Fluted (grooved) roller — the standard Indian seed-drill feed	Slide the roller to change the EXPOSED FLUTE LENGTH (and change speed via drive sprockets)
Internal double-run feed	DRIVE RATIO (gears/sprockets) and choice of the large/small side
Cup feed	Cups on a spindle; spindle speed
Fertilizer metering (star wheel/rotor/gravity agitator)	Calibrated identically, by weight

- **Uniformity:** collect from EACH opener separately. The coefficient of variation (CV) between openers shows metering uniformity — investigate tubes/rollers if any opener deviates markedly.

Field vs laboratory rate — wheel slip

- In the field the ground wheel slips/skids (typically **5–7%**): it makes FEWER revolutions per actual distance travelled, so the drill meters **LESS** seed per hectare than the stationary calibration suggests.
- Field check: sow a known area and weigh the seed actually used — or add a slip correction.
- Modern drills with electronic drives use radar/GPS speed and avoid this error.

Practice MCQs

Q1. Calibration of a seed drill is done to:

(a) Increase draft
(b) Determine and set the actual seed rate before field sowing
(c) Test the frame strength
(d) Measure fuel use

Q2. Area covered by a drill of width W in N revolutions of a wheel of diameter D is:

(a) πDNW
(b) DNW
(c) πD²NW/4
(d) 2πDNW

Q6. Seed rate of a fluted-roller drill is adjusted mainly by:

(a) Tyre pressure
(b) Exposed length of the flutes
(c) Hopper size
(d) Furrow opener angle

Q7. In an internal double-run feed, seed rate is changed chiefly by:

(a) Drive gear/sprocket ratio
(b) Painting the rotor
(c) Seed treatment
(d) Hitch height

Q8. The most common seed-metering device in Indian seed drills is:

(a) Vacuum disc
(b) Fluted roller
(c) Finger pickup
(d) Belt

Q9. Calibration must use the actual seed lot because rate depends on:

(a) Seed size, shape, density and coating
(b) Field slope
(c) Tractor make
(d) Moon phase

Q10. Collecting seed separately from each opener during calibration checks:

(a) Draft
(b) Metering uniformity between openers (CV)
(c) Wheel slip
(d) Tank leaks

Q11. Because of drive-wheel slip (5–7%), the field seed rate compared with stationary calibration is:

(a) Higher
(b) Lower
(c) Same
(d) Double

Q35. Before counting calibration output, the metering mechanism is first run a few turns to:

(a) Warm the bearings
(b) Fill/prime the feed mechanism for steady metering
(c) Polish the rollers
(d) Check tyres

Answers

Q1 — (b) Determine and set the actual seed rate before field sowing
Q2 — (a) πDNW
Q6 — (b) Exposed length of the flutes
Q7 — (a) Drive gear/sprocket ratio
Q8 — (b) Fluted roller
Q9 — (a) Seed size, shape, density and coating
Q10 — (b) Metering uniformity between openers (CV)
Q11 — (b) Lower
Q35 — (b) Fill/prime the feed mechanism for steady metering

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 11 (Crop Planting) & Ch. 21 (Spraying and Dusting); Jagdishwar Sahay — Elements of Agricultural Engineering; ISI/BIS & RNAM test codes for seed drills and sprayers; NPTEL Farm Machinery lectures.